CLARKSON UNIVERSITY
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

EE412
Senior Design

Fall 2026 | 3 credits | Dr. William Jemison

Fall 2026 senior-design format

EE412 is the Electrical Engineering capstone. The course uses a common, structured design process for all teams while allowing a broad range of technically substantial projects. A project may emphasize power and energy, electronics, instrumentation, sensing, controls, communications, signal processing, RF, embedded systems, robotics, biomedical/environmental applications, AI/ML-enabled engineering systems, or another approved area. The required design evidence is common even when the technical implementation is very different.

Course at a Glance

Course	EE412 Senior Design, 3 credits, Fall 2026
Instructor / Coordinator	Dr. William Jemison, Professor, Electrical & Computer Engineering
Email	wjemison@clarkson.edu
Phone	315-268-6509
Office / office hours	See Moodle / Clarkson directory; meetings by appointment as needed
Class meeting	Tuesday, Thursday 8:00–9:30 a.m., SC 340
Lab work	Tuesday, Thursday 12:30–4:15 p.m., CAMP 195
Term	August 24–December 4, 2026; final-examination period December 7–11
Project work	Additional project-team, laboratory, sponsor, or faculty-mentor meetings are required as appropriate to the project and may occur outside the common class period
Course systems	Moodle for announcements, rubrics, grades, and official submissions; an instructor-approved shared repository for project management, design files, notebooks, data, and retained evidence
Final requirement	Validated capstone system/design, final demonstration, final report, oral presentation, technical poster/summary, and complete archival evidence package

How to communicate

Use your Clarkson email account and include **EE412** and your team/project name in the subject line. Technical, safety, equipment, sponsor, or team-functioning concerns should be raised promptly. Teams should not wait for a milestone review to disclose a problem that could affect safety, ethics, schedule, scope, data integrity, or project completion.

Roles in the course

Dr. Jemison serves as course instructor/coordinator and is responsible for common course expectations, milestone structure, professional-practice instruction, ABET-aligned assessment, and final course grades. Projects may also have a **faculty mentor, laboratory supervisor, industry/community sponsor, or technical advisor**. Those mentors provide project-specific technical guidance, but the student team remains responsible for engineering decisions, documentation, testing, and meeting EE412 requirements.

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1 Course Purpose, Project Scope, Preparation, and Materials

1.1 Course description

EE412 is the culminating Electrical Engineering design experience. Students work in teams on one or more approved engineering design projects involving problem definition, planning, analysis, modeling or simulation, preliminary and detailed design, implementation or construction when appropriate, experimentation, testing, evaluation, documentation, demonstration, and oral communication. The course emphasizes a complete engineering solution rather than a single analysis, isolated experiment, or collection of loosely connected tasks.

1.2 Common class versus project-laboratory time

The classroom/Lab meeting is used for project-management instruction, requirements and design methods, professional practice, safety/ethics/standards, design reviews, short presentations, milestone coordination, and team check-ins. The project itself normally requires substantial work outside this period in the appropriate laboratory or project environment. Because projects may occur in different laboratories, milestone demonstrations may be scheduled by team appointment when a common meeting period is not sufficient.

1.3 Broad project scope

EE412 supports a wide variety of senior-design projects. Appropriate projects may include, but are not limited to:

- electric power, renewable energy, power electronics, machines, smart-grid or energy systems;
- analog, mixed-signal, digital, RF/microwave, communications, instrumentation, or measurement systems;
- controls, automation, robotics, mechatronics, autonomous or cyber-physical systems;
- sensors, biomedical, environmental, electrochemical, health, infrastructure, or industrial monitoring systems;
- signal processing, audio, imaging, radar, localization, communications, or data-acquisition systems;
- embedded/IoT systems, edge computing, or AI/ML-enabled engineering systems when the project includes defensible engineering requirements, design trade-offs, validation, and realistic constraints;
- computational, simulation-intensive, or algorithmic designs when the final work constitutes a complete engineering design and is validated against quantitative requirements;
- interdisciplinary projects in which Electrical Engineering contributes a substantial, identifiable design component.

Note

Not every EE412 team must build the same kind of prototype, use the same hardware, or follow the same technical workflow. However, every project must show the same capstone design discipline: measurable requirements, alternatives and trade-offs, standards and constraints, planning, detailed design, implementation or realization, appropriate experimentation/testing, evidence-based validation, professional/ethical judgment, teamwork, and communication.

A literature review, a preselected implementation with no design choices, routine assembly of a kit, coding without engineering requirements, data collection without a design question, or research exploration without a defined engineering solution is not by itself sufficient for the capstone. A research-oriented project is acceptable when the team converts the research need into a design problem with stakeholders, requirements, alternatives, constraints, a testable deliverable, and a defensible verification/validation plan.

1.4 Prerequisite and expected preparation

Prerequisite: EE311. Students are expected to apply knowledge acquired across the Electrical Engineering curriculum and to independently learn project-specific material when needed.

Students should be prepared to:

- formulate quantitative, testable requirements from an open-ended need;
- apply circuit, electronics, power, signals, controls, communications, computation, instrumentation, or other EE knowledge appropriate to the project;

- use modeling, simulation, calculation, prototyping, programming, CAD, laboratory instrumentation, and data analysis as appropriate;
- plan and conduct safe experiments or tests and record configuration, units, calibration, uncertainty, limitations, and raw data;
- locate and interpret authoritative standards, regulations, datasheets, technical literature, and manufacturer documentation;
- compare feasible alternatives and explain trade-offs using engineering evidence rather than preference alone;
- communicate technical work in a form that another qualified engineer could understand, evaluate, reproduce, and continue.

1.5 Instructional materials

There is no required textbook. The instructor will provide or link readings on engineering design, project planning, standards, ethics, safety, experimentation, technical communication, and assessment. Teams must also use project-specific authoritative references, such as IEEE/IEC/ISO/NIST/ANSI standards, OSHA or other safety guidance when applicable, component datasheets, peer-reviewed literature, codes, utility/industry requirements, and sponsor documentation.

A useful optional design reference is Ralph M. Ford and Chris S. Coulston, *Design for Electrical and Computer Engineers*, McGraw–Hill, 2005.

1.6 Project repository and engineering record

Each team must maintain an instructor-approved shared repository containing, as applicable:

- requirements, specifications, architecture, schematics, models, drawings, interface definitions, simulations, calculations, and design decisions;
- source code/configuration files, bill of materials, procurement records, instrument settings, and version information;
- a dated individual engineering notebook for each student;
- raw and processed data, analysis scripts, calibration information, photographs/video, and test evidence;
- WBS/task ownership, Gantt chart, risk register, budget, issue/change log, and meeting/action records;
- standards/constraints matrix, ethical/professional analysis, experimentation/test plans, and requirements-to-evidence traceability;
- milestone packages and final communication/archive materials.

2 Learning Outcomes and ABET-Aligned Assessment Evidence

EE412 is a major design experience for the Electrical Engineering program. The course is structured so that the same core assessment evidence can be retained across technically different projects. The project must draw on prior coursework, require engineering judgment and iteration, incorporate appropriate engineering standards, address realistic constraints, and produce credible evidence that the final solution meets its intended need.

2.1 Course learning outcomes

By the end of the semester, each student will be able to:

CO	Student capability	Primary ABET alignment
CO1	Define an open-ended engineering problem; translate stakeholder needs into measurable requirements; generate alternatives; and apply engineering design to produce a solution under realistic constraints.	SO2; supports SO1

CO	Student capability	Primary ABET alignment
CO2	Identify and apply appropriate engineering standards, regulations, safety practices, and constraints, and justify major design decisions through documented engineering trade-offs.	SO2; supports SO4
CO3	Recognize ethical and professional responsibilities and make informed judgments considering public health, safety, welfare, global, cultural, social, environmental, and economic impacts.	SO4
CO4	Function effectively on a team by establishing goals and roles, planning tasks, sharing responsibility, supporting inclusive collaboration, resolving problems, and meeting objectives.	SO5
CO5	Develop and conduct appropriate experimentation, verification, and validation; analyze and interpret data; account for uncertainty and limitations when relevant; and use engineering judgment to draw conclusions.	SO6
CO6	Acquire and apply new knowledge using appropriate learning strategies, authoritative resources, and independent technical investigation.	SO7
CO7	Communicate engineering requirements, design rationale, evidence, limitations, and recommendations effectively through written reports, oral presentations, visual material, and demonstrations.	SO3 (course-level evidence)

2.2 Primary program-assessment outcomes

For continuity with the Electrical Engineering program's recent course-level assessment, EE412 is designed to retain especially clear direct evidence for **ABET Student Outcomes 2, 4, 5, 6, and 7**. Communication (SO3) and technical problem solving (SO1) are strongly exercised in the course even when they are not selected as the primary outcomes for a particular program-assessment cycle.

SO	Core capability	Primary retained EE412 evidence
2	Engineering design under realistic constraints	Requirements/specification; standards/constraints matrix; alternatives/trade study; detailed design; design review; final validation against requirements
4	Ethical/professional judgment and impacts	Individual professional-practice paper(s); safety/impact analysis; ethics case analysis; final Ethical and Professional Responsibilities section; oral defense
5	Team effectiveness	Team contract; WBS/roles; Gantt chart; CATME/instructor peer evidence; milestone contribution records; team reflection
6	Experimentation and engineering judgment	Test/experiment plans; instrument/configuration records; raw data; analysis; uncertainty/limitations; failed-test/corrective-action evidence; final validation
7	New knowledge and learning strategies	Individual engineering notebook; annotated references; new-learning memo/section; use of standards/literature; design-review defense of newly learned material

2.3 Common ABET design artifacts across all projects

Regardless of project topic, every team must maintain the following common evidence:

1. **Requirements and Success-Criteria Matrix** with stable requirement IDs and measurable acceptance criteria.
2. **Standards, Regulations, Safety, and Realistic Constraints Matrix** connecting each applicable item to design decisions and verification evidence.

3. **Alternatives and Engineering Trade-off Record** showing at least one consequential decision for which multiple feasible approaches were compared.
4. **Requirements-to-Evidence Traceability Matrix** mapping each requirement to analysis, simulation, inspection, demonstration, test, or other validation evidence.
5. **Risk and Safety Register** including technical, schedule, procurement, experimental, human, cybersecurity/privacy, and other relevant risks.
6. **Individual Engineering Notebooks and Contribution Evidence** sufficient to distinguish individual from team work.
7. **Experiment/Test Record** preserving procedure, equipment/configuration, raw data, analysis, failures, corrective actions, limitations, and conclusions.

2.4 Standards and constraints requirement

The Standards, Regulations, Safety, and Realistic Constraints Matrix begins in Milestone 1 and is revised through the final report. Teams shall address what is applicable rather than forcing irrelevant categories. Potential categories include:

- electrical and laboratory safety, high voltage/current, rotating machinery, battery, laser, RF, chemical/biological, thermal, mechanical, and field-deployment hazards;
- IEEE, IEC, ISO, ANSI, NIST, NFPA, NEC, OSHA, FCC, utility, communications, accessibility, cybersecurity, privacy, or domain-specific standards and regulations;
- cost, schedule, supply-chain availability, manufacturability, maintainability, reliability, size, weight, power/energy, bandwidth, accuracy, speed, environmental conditions, sustainability, usability, accessibility, security/privacy, public health, and societal impact;
- sponsor requirements and Clarkson laboratory/IT/research policies.

ABET evidence principle

A standard, ethical concern, safety issue, or realistic constraint should not appear only as a paragraph in the final report. The assessment evidence should show how it changed or justified a requirement, design choice, test, risk mitigation, operating procedure, or conclusion.

2.5 Engineering trade-off expectation

At least one major design decision must compare two or more technically feasible alternatives. The comparison should use quantitative criteria when possible and may include performance, cost, safety, efficiency, reliability, uncertainty, implementation risk, schedule, maintainability, environmental impact, accessibility, or other project-specific factors. The team must explain why the selected design is appropriate and what was sacrificed by not choosing the alternatives.

2.6 Experimentation and validation expectation

Every project must include appropriate verification/validation evidence. The exact method may differ by project: laboratory measurement, field test, simulation-to-measurement comparison, hardware-in-the-loop test, controlled experiment, statistical performance evaluation, power-flow study, communications test, controller performance test, RF measurement, algorithmic benchmark, or another defensible method. The final conclusion must be based on data and engineering judgment rather than demonstration alone.

2.7 Four-level ABET assessment rubric

Selected retained artifacts will be scored independently of the course letter grade using the department's four-level language:

- **Exceeds Expectations**: evidence is complete, technically sound, well justified, and demonstrates independent engineering judgment beyond the minimum expected level;
- **Meets Expectations**: evidence is substantially complete and correct and demonstrates the intended student outcome at the graduating-student level;

- **Needs Improvement**: important evidence is incomplete, weakly justified, or contains technical/professional deficiencies that limit attainment;
- **Urgent to Improve**: evidence is absent, seriously deficient, unsafe/unethical, or does not demonstrate the intended outcome.

Where useful, the instructor may map rubric scores or percentages to these categories for internal assessment. Course grades and ABET attainment are related but are not identical measures.

2.8 Individual versus team evidence

Because capstone work is team-based, direct program assessment will use both team artifacts and individual evidence. An individual may be asked to explain requirements, standards, safety, trade-offs, experiments, data, design decisions, or ethical implications during a milestone/design review. CATME/peer evaluation, notebooks, progress reports, individual position papers, and individual design-review performance may be used to determine whether a student personally demonstrates the assessed outcome.

3 Project Organization, Milestones, and Deliverables

3.1 Project selection and approval

Projects may originate from ECE faculty, other Clarkson faculty, industry/community partners, student proposals, research laboratories, competitions, or ongoing sponsored work. The instructor approves the capstone scope. Approval depends on the presence of a substantive Electrical Engineering design component, feasible semester scope, measurable deliverables, appropriate technical challenge, access to needed facilities/resources, and enough design freedom for students to make and justify consequential engineering decisions.

A project may be interdisciplinary, but each EE412 student must have identifiable Electrical Engineering responsibilities and must be able to demonstrate the EE412 learning outcomes through their own work.

3.2 Teams, roles, and project control

Teams must establish a decision process, meeting schedule, communication norms, escalation path, code of ethical conduct, and task-ownership structure. Every major task must have a primary owner and a backup. Teams shall maintain a WBS, Gantt chart, risk register, budget/procurement status, decision log, and individual notebooks. Plans may change; undocumented changes are not project management.

3.3 Common milestone submission rules

Unless a rubric states otherwise, each milestone package must:

- be submitted by **5:00 p.m.** on the listed due date through the designated system;
- use the instructor-provided report structure/template and maintain revision history and contribution statements;
- preserve stable IDs for requirements, tests/evidence, risks, and major design decisions;
- distinguish planned work, completed work, measured results, assumptions, limitations, failures, and future work;
- include links/references to notebooks, drawings, calculations, simulations, source code, raw data, photographs/video, instrument settings, or other reproducible evidence;
- carry forward the current requirements matrix, standards/constraints matrix, risk register, schedule/WBS, budget, and requirements-to-evidence traceability;
- identify external assistance, reused designs/code, licenses, sponsor contributions, and generative-AI assistance as required by course policy.

3.4 Milestone 0: Project definition and team launch

Project/team confirmation by Friday, September 4, 2026.

Each team completes:

1. project title, sponsor/faculty mentor, short problem statement, preliminary scope, and intended final deliverable;
2. team contract, code of ethics, roles, communication plan, and initial task ownership;
3. preliminary resource/facility needs, sponsor restrictions, safety concerns, data/privacy/IP considerations, and procurement concerns;
4. initial one-page project charter with the reasons the project qualifies as a senior design experience.

3.5 Milestone 1: Needs, requirements, standards, and validation concept

Due Friday, September 11, 2026, at 5:00 p.m.

The Milestone 1 package includes:

1. stakeholder/problem statement, operating context, assumptions, exclusions, intended users, and measurable success criteria;
2. uniquely numbered functional and nonfunctional requirements with measurable acceptance criteria;
3. preliminary system/context diagram and interface inventory;
4. Preliminary Standards, Regulations, Safety, and Realistic Constraints Matrix;

5. preliminary stakeholder/impact analysis covering public health, safety, welfare, environmental, economic, societal, accessibility, security/privacy, and ethical issues as applicable;
6. draft final verification/validation plan and requirements-to-evidence traceability matrix;
7. initial risk register, deliverables list, procurement/resource plan, and project schedule framework;
8. individual engineering-notebook evidence of each student's contribution.

No functional prototype is required at this milestone. The emphasis is on whether the team has defined the right engineering problem and can state how success will be measured.

3.6 Milestone 2: Concept development, alternatives, and detailed project plan

Due Friday, September 25, 2026, at 5:00 p.m.

The package includes:

1. high-level system architecture or design concept, including major electrical, computational, physical, user, and external-system elements as applicable;
2. at least one **Engineering Trade-off Analysis** comparing two or more feasible alternatives for a consequential design decision;
3. preliminary models/calculations/simulations that support feasibility;
4. interface definitions, major design assumptions, requirement allocation, and preliminary design margins;
5. detailed WBS with primary and backup owners, dependencies, a Gantt chart, milestone goals, and critical schedule risks;
6. preliminary BOM/budget/procurement plan;
7. revised requirements, standards/constraints matrix, risk register, and verification plan;
8. a short **Technology Serving Humanity / Stakeholder Impact Assessment** describing benefits, possible adverse effects, foreseeable misuse/failure, accessibility, sustainability, and appropriate safeguards;
9. demonstration or evidence of readiness with the principal design, simulation, development, fabrication, or test tools needed for the project.

3.7 Milestone 3: Detailed design and experiment/test plan

Due Friday, October 9, 2026, at 5:00 p.m.

The package includes:

1. detailed design documentation sufficient for an appropriately prepared engineering student to implement or reproduce the planned work;
2. schematics, drawings, models, algorithms, control diagrams, state/sequence diagrams, equations, simulations, interface specifications, component selection, tolerances/margins, and design calculations as applicable;
3. evidence that major design choices have been checked against standards, safety, realistic constraints, and stakeholder needs;
4. a detailed experiment/verification plan with equipment, setup, calibration, variables, data to collect, sample size/repetitions where appropriate, uncertainty/limitations, expected results, and pass/fail criteria;
5. updated schedule, budget, procurement status, risks, and requirements-to-evidence traceability;
6. preliminary implementation/prototype/simulation/test-fixture evidence appropriate to the project;
7. CATME or instructor-designated peer evaluation.

3.8 Milestone 4: Implementation/prototype and subsystem verification

Due Friday, October 23, 2026, at 5:00 p.m.

The package includes:

1. implemented hardware/software/model/prototype or other project realization appropriate to the approved scope;

2. subsystem or intermediate verification evidence using the approved procedures;
3. raw data/configuration, pass/fail results, anomalies, failed tests, root-cause analysis, and corrective actions;
4. updated detailed design for remaining work and an integration plan;
5. revised standards/constraints compliance, safety/risk assessment, requirements traceability, schedule, budget, and decision record;
6. a working demonstration, measured result, or equivalent evidence of substantive technical progress;
7. **individual design review**: written and oral explanation of the student's major design responsibility, including requirements, alternatives, calculations/models, standards/constraints, risk/safety, test strategy, and current evidence;
8. CATME or instructor-designated peer evaluation.

3.9 Milestone 5: Integrated solution and final-validation readiness

Due Friday, November 6, 2026, at 5:00 p.m.

The package includes:

1. an integrated system/design demonstrating the planned core functionality or equivalent validated design state;
2. completed intermediate testing and a revised final verification/validation plan;
3. complete requirements-to-evidence traceability matrix identifying remaining evidence gaps;
4. performance, robustness, stress, fault/failure, sensitivity, or uncertainty testing as appropriate to the project;
5. updated standards/constraints compliance, safety assessment, risk register, budget, schedule, configuration/release plan, and known-issues list;
6. a final technical demonstration/readiness review;
7. CATME or instructor-designated peer evaluation.

3.10 Final validation and capstone communication package

From Milestone 5 through the end of the semester, the team executes the approved final validation plan, closes traceability gaps, documents failures as well as successes, improves the design where evidence justifies iteration, and prepares the final professional package.

The final package includes:

- final validated design/system and demonstration;
- complete requirements-to-evidence traceability and final standards/constraints compliance status;
- final report with design rationale, experiments/tests, data, results, limitations, lessons learned, and recommendations;
- dedicated **Ethical and Professional Responsibilities** section addressing the relevant SO4 considerations and mitigation decisions;
- final oral presentation with participation by each team member;
- technical poster or equivalent concise visual summary suitable for a design showcase/RAPS-style audience;
- complete archive containing design files, code/configuration, data, analysis, notebooks, BOM, user/operating guidance, and release/reproducibility information.

Final performance is requirements-based

A visually impressive demonstration does not replace validation. The final project is evaluated primarily by how well the team has defined the need, justified the design, managed realistic constraints, conducted appropriate experimentation/testing, and produced evidence that the solution satisfies its requirements safely and professionally.

4 Assessment and Grading

4.1 Assessment philosophy

EE412 combines team engineering with individual accountability. Team deliverables are appropriate because engineering design is collaborative, but an individual student's grade may be adjusted when contribution evidence, peer evaluation, notebooks, design reviews, or oral questioning show substantially different levels of participation or understanding.

The course uses a 1000-point structure so that design process, final performance, experimentation, professional practice, communication, and individual contribution are all visible rather than collapsed into a single "project accomplishment" grade.

4.2 Point breakdown

Component	Score type	Points
Milestone packages/reviews (M1–M5, 50 points each)	Team shared, individually moderated	250
Final validated project/design and capstone technical assessment	Team shared, individually moderated	200
Individual design review: written report	Individual	50
Individual design review: oral presentation/defense	Individual	50
Engineering notebook and individual progress evidence	Individual	100
Professional-practice / ethics / standards position papers	Individual	100
Final team report and technical archive	Team shared	100
Final oral presentation	Team shared + individual speaking evidence	50
Poster / concise technical communication artifact	Team shared	50
Peer contribution and instructor professional evaluation	Individual	50
Total		**1000**

4.3 Professional-practice position papers

To retain strong individual ABET evidence for SO4 and SO7, students complete short individual professional-practice assignments during the semester. Topics may include:

1. an engineering ethics/safety/standards judgment connected to the student's own project; and
2. analysis of an engineering failure, incident, or professional dilemma, including consequences for public safety, environment, economics, stakeholders, and the student's future engineering practice.

The exact prompts and rubrics will be posted in Moodle. The papers are individual work even when they discuss the team project.

4.4 Milestone scoring

Each 50-point milestone normally evaluates five dimensions:

- engineering completeness and technical quality;
- evidence and traceability;
- planning, risk, budget, and project control;
- standards, safety, constraints, and professional considerations;
- demonstration/review quality and team readiness.

Rubrics may change the weighting by milestone. A milestone package that is incomplete or technically unsound may require correction/resubmission so that later design work does not build on an unresolved error. Resubmission normally improves the project record but does not automatically replace the original grade unless the instructor states otherwise.

4.5 Individual contribution adjustment

A team score assumes approximately equitable contribution. The instructor may adjust an individual's share of team points using engineering notebooks, progress records, repository history, CATME/peer evaluation, milestone questioning, sponsor/faculty-mentor input, attendance/participation, and demonstrated knowledge. A student should never claim work in a progress report that cannot be supported by retained evidence.

4.6 Final capstone technical assessment

The 200-point final technical assessment is not based only on whether the system "works." It evaluates:

- satisfaction of approved requirements and success criteria;
- technical quality and integration of the design;
- use of appropriate standards, realistic constraints, and safety practices;
- quality of experiments/tests and the credibility of the data and conclusions;
- robustness, limitations, known failures, and engineering response to unexpected results;
- quality of final configuration, reproducibility, user/operating guidance, and project handoff.

4.7 Suggested grading thresholds

Total points	Approximate letter grade
970–1000	A+
930–969	A
900–929	A-
870–899	B+
840–869	B
800–839	B-
760–799	C+
700–759	C
660–699	C-
600–659	D
Below 600	F

Final thresholds may be adjusted by the instructor consistent with Clarkson grading policies; students will be informed of any material change.

4.8 ABET evidence-to-assessment map

SO	Primary direct evidence	Secondary corroborating evidence
2	M1 requirements/constraints; M2 trade study; M4 individual design review; final traceability/validation	M3 detailed design; M5 compliance/readiness; final report
4	Individual professional-practice paper(s); final Ethical and Professional Responsibilities section	Standards/safety matrix; milestone questioning; instructor observation
5	CATME/peer evidence; team contract/WBS; final team contribution reflection	Milestone contribution statements; instructor observation; project-management records
6	M3 experiment/test plan; M4 verification data; final validation evidence	Notebooks; design review; analysis scripts; uncertainty/limitations discussion
7	Individual notebook/new-learning evidence; professional-practice research; design-review defense	References/standards used in milestones; final report prior-knowledge/new-learning discussion

5 Fall 2026 Schedule and Major Due Dates

Regular meeting pattern

Tuesday, Thursday 8:00–9:30 a.m., SC 340 for Classroom meeting and 12:30–4:15 p.m., CAMP 195 for lab work. Project teams also meet in their assigned laboratories/project spaces and with mentors/sponsors as needed. Moodle is the authoritative source for team-specific milestone review and demonstration appointments.

Week / dates	Anchor	Planned focus and major deliverables
Aug. 24–28	Week 1	Course launch; capstone expectations; project briefings; project selection; introduction to ABET evidence, engineering design, team formation, and project charter.
Aug. 31–Sept. 4	Week 2	Team organization; ethics; stakeholder needs; safety and project feasibility. **Milestone 0 / team contract and project confirmation due Fri. Sept. 4.**
Sept. 7–11	Week 3	Requirements engineering; measurable success criteria; standards/constraints; verification concepts. **Milestone 1 due Fri. Sept. 11 at 5 p.m.**
Sept. 14–18	Week 4	Architecture/concept generation; engineering alternatives; modeling/feasibility. Career Fair: Tues. Sept. 15.
Sept. 21–25	Week 5	Trade studies; WBS/Gantt; procurement; risk; Technology Serving Humanity / impact analysis. **Milestone 2 due Fri. Sept. 25 at 5 p.m.**
Sept. 28–Oct. 2	Week 6	Detailed design methods; engineering calculations/simulation; experiment and test planning.
Oct. 5–9	Week 7	Detailed design completion; measurement strategy; uncertainty/limitations. **Milestone 3 due Fri. Oct. 9 at 5 p.m.**
Oct. 12–16	Week 8	**Fall Recess Mon. Oct. 12–Tues. Oct. 13; classes resume Wed. Oct. 14.** Implementation/prototyping; individual design-review preparation.
Oct. 19–23	Week 9	Subsystem verification; design reviews; failed-test/root-cause analysis; integration planning. **Milestone 4 due Fri. Oct. 23 at 5 p.m.**
Oct. 26–30	Week 10	Integration; professional-practice/ethics case discussion; validation planning; project recovery where needed.
Nov. 2–6	Week 11	Integrated solution; final test readiness; requirements-to-evidence gap review. Spring enrollment begins Wed. Nov. 4. **Milestone 5 due Fri. Nov. 6 at 5 p.m.**
Nov. 9–13	Week 12	Final testing; data analysis; poster/technical communication development. **Fall RAPS: Fri. Nov. 13.** Teams may use a RAPS-style poster/showcase opportunity when appropriate.
Nov. 16–20	Week 13	Final validation; standards/constraints closure; final report drafting; professional/ethical responsibility evidence; final project readiness review.
Nov. 23–27	Week 14	Final report/analysis work early in week. **Thanksgiving Recess Wed. Nov. 25–Fri. Nov. 27.** No common class meetings during recess.
Nov. 30–Dec. 4	Week 15	Classes resume Mon. Nov. 30. Final demonstrations/presentations as scheduled; final report revision and archive preparation. **Last day of class: Fri. Dec. 4.**
Dec. 7–11	Finals	Final-examination period. Final presentation/demo slot and/or final revised archive due according to the Registrar/Moodle schedule.

5.1 Typical recurring expectations

Unless the instructor announces otherwise:

- individual engineering notebooks are maintained continuously and reviewed at milestones;
- teams update WBS/Gantt, risks, budget/procurement, and requirements-to-evidence traceability at least weekly;
- short progress/status updates may be requested weekly in Moodle or the designated project repository;
- individual design reviews are scheduled during the Milestone 3–4 portion of the semester when a student's design is mature enough to defend but before final implementation is locked;
- team-specific demonstrations may be scheduled outside the common Tu/Tr period when laboratory equipment or extended test time is required.

6 Course Policies and Professional Practice

6.1 Attendance and participation

Regular participation in the common class and required project meetings/reviews is expected. Senior design depends on team coordination and access to shared facilities, sponsors, and mentors. If a student must miss a required activity, the student should notify the instructor/team as early as possible and arrange a reasonable recovery plan. Repeated absence, missed design reviews, or failure to participate in team work may affect individual contribution scores.

6.2 Laboratory and project safety

Safety is part of engineering design and is assessed, not treated as an administrative afterthought. Teams must identify hazards before construction/testing and follow the rules of the laboratory or field environment in which they work. Projects involving high voltage/current, batteries, rotating machinery, pressure, chemicals, lasers, RF transmission, heat, biological/health data, human participants, vehicles, field deployment, or other elevated hazards require appropriate review and mitigation before testing.

Stop work and contact the instructor/faculty mentor when a task presents an unreviewed hazard, requires authorization the team does not have, or would violate a laboratory, University, sponsor, legal, or professional requirement.

6.3 Academic integrity and attribution

Students are responsible for complying with Clarkson's Academic Integrity policy. Engineering work must distinguish original student contributions from reused code, reference designs, published circuits, vendor examples, prior team work, sponsor-provided material, AI assistance, and other external contributions. Sources must be cited, and licensing/IP restrictions must be respected.

Fabrication or alteration of measurements, simulation results, experiment records, notebook history, references, peer contributions, or test evidence is a serious academic and professional violation.

6.4 Generative AI and external computational tools

Generative AI and similar tools may be used to support learning, brainstorming, coding assistance, editing, troubleshooting, or documentation when the use is consistent with the assignment and is transparent. Students remain fully responsible for correctness, originality, safety, licensing, confidentiality, and all submitted claims.

Students must:

- independently verify AI-generated technical claims against authoritative sources, calculations, compilation, simulation, or measured evidence;
- not upload confidential, proprietary, export-controlled, personally identifiable, sponsor-restricted, or security-sensitive information to an external AI service without authorization;
- not use AI to fabricate data, measurements, references, images of purported results, notebooks, or completion evidence;
- disclose substantial AI assistance in an **AI and External Assistance Statement** identifying the tool, purpose, affected work, and human verification performed;
- be prepared to explain and defend any AI-assisted design, code, analysis, or writing during an oral review.

The instructor may restrict AI use for a specific assessment when independent performance is required.

6.5 Teamwork, CATME, and conflict resolution

Teams should use the team contract and project records to address problems early. Concerns should be specific and evidence-based: missed tasks, unsafe work, poor communication, exclusion, unprofessional conduct, or inaccurate contribution claims should be documented and discussed with the instructor when normal team resolution is unsuccessful. CATME or another instructor-designated peer-assessment tool may be used during the semester and at the end of the project.

Peer evaluation is not a popularity score. Ratings must reflect contribution, reliability, communication, leadership/followership, preparation, and support of team objectives. Retaliation, collusion, or intentionally misleading ratings are unacceptable.

6.6 Engineering notebooks and reproducibility

Each student maintains a dated engineering notebook. It should contain enough evidence to establish what the student did, when decisions/tests occurred, what sources were used, what failed, and how the work evolved. The notebook may include calculations, sketches, design rationale, instrument settings, photographs, data links, simulation results, code references, meeting decisions, and technical reflections. A final polished report cannot substitute for missing contemporaneous evidence.

6.7 Human data, privacy, cybersecurity, and public release

Projects that collect audio, video, biometric, health, location, network, or other potentially sensitive data must use appropriate consent/approval, minimize data collection, protect credentials and personal information, define storage/retention, and comply with Clarkson/sponsor requirements. Research involving human participants must not begin without any required institutional approval.

Public release through GitHub, arXiv, YouTube, posters, demonstrations, or external repositories must be reviewed for intellectual property, sponsor restrictions, licensing, privacy, security, export controls, and accessibility. When public release is inappropriate, the instructor will designate a controlled archive.

6.8 Professional communication and accessibility

Students are expected to communicate respectfully and professionally with teammates, instructors, mentors, sponsors, technicians, and other stakeholders. Technical disagreement should be based on evidence and should not become personal. Reports, posters, figures, videos, and presentations should be legible and accessible; spoken video content should be captioned when required, and visual material should use clear labels and readable contrast.

6.9 Late work and schedule recovery

Senior design is milestone-driven because downstream work depends on timely decisions. When circumstances beyond a team's control threaten a deadline, the team should notify the instructor before the deadline when possible and submit a written recovery plan. An extension does not remove the responsibility to update the Gantt chart, risk register, task ownership, procurement plan, or sponsor expectations.

6.10 Institutional policies and accommodations

Students are responsible for current University policies, including academic integrity, attendance/absence requirements, accessibility/accommodations, nondiscrimination/harassment, and religious accommodations. Students who need academic accommodations should work with Clarkson's Office of Accessibility Services and discuss implementation with the instructor as early as possible, especially when accommodations affect laboratory safety, field work, team scheduling, demonstrations, or presentation formats.

Useful official sources include:

- Clarkson Regulations;
- Clarkson Academic Calendar;
- Office of Accessibility Services;
- Non-Discrimination resources;
- Dean of Students / student support resources.

Student acknowledgment

By participating in EE412, each student accepts responsibility for reading the syllabus and milestone rubrics; maintaining accurate individual evidence; following applicable safety, ethical, legal, and professional requirements; protecting restricted information; contributing fairly to team work; and asking for clarification before acting when a requirement is uncertain.